

AN IRRATIONAL FIXED POINT FOR A SALEM BETA-TRANSFORMATION: A SCOPE CORRECTION

ARGUS OPEN RESEARCH PROJECT

ABSTRACT. For the beta-transformation $T_\beta(x) = \beta x \pmod{1}$ on $[0, 1)$, consider

$$p(t) = t^4 - 3t^3 + 3t^2 - 3t + 1.$$

We prove that the larger real root β of p is a Salem number greater than two and that

$$x = \frac{1}{\beta - 1}$$

is an irrational fixed point of T_β . Hence the literal identity $\text{Per}(\beta) = \mathbb{Q} \cap [0, 1)$ is false. This does not address the classical Salem-base problem formulated by Schmidt, whose right-hand side is $\mathbb{Q}(\beta) \cap [0, 1)$; indeed the displayed fixed point belongs to $\mathbb{Q}(\beta)$. The result is therefore a correction of an over-restricted restatement, not a resolution of Schmidt's open problem.

1. STATEMENT AND SOURCE DISTINCTION

For a real number $\beta > 1$, define

$$T_\beta: [0, 1) \longrightarrow [0, 1), \quad T_\beta(x) = \beta x \pmod{1}.$$

Following Schmidt [1], let $\text{Per}(\beta)$ be the set of points whose forward T_β -orbit is finite. This is the same as eventual periodicity of the beta-expansion.

A catalogue restatement asserted the following identity for every Salem number:

$$\text{Per}(\beta) = \mathbb{Q} \cap [0, 1). \tag{1}$$

The source does not state (1). Schmidt proves the corresponding identity with $\mathbb{Q}(\beta)$ for Pisot numbers and explicitly leaves open, for Salem numbers, the question

$$\text{Per}(\beta) = \mathbb{Q}(\beta) \cap [0, 1). \tag{2}$$

Theorem 1.1. *Let β be the larger real root of*

$$p(t) = t^4 - 3t^3 + 3t^2 - 3t + 1.$$

Then $\beta > 2$ is a Salem number, and

$$x = \frac{1}{\beta - 1}$$

belongs to $\text{Per}(\beta) \setminus \mathbb{Q}$. Consequently (1) is false. The point x belongs to $\mathbb{Q}(\beta)$, so the construction is not a counterexample to (2).

2. IRREDUCIBILITY

Lemma 2.1. *The polynomial p is irreducible over $\mathbb{Q}$.*

Proof. Reduction modulo two gives

$$\overline{p}(t) = t^4 + t^3 + t^2 + t + 1 \in \mathbb{F}_2[t].$$

It has no root in $\mathbb{F}_2$, since $\overline{p}(0) = \overline{p}(1) = 1$. A reducible root-free quartic over $\mathbb{F}_2$ would have an irreducible quadratic factor. The only monic irreducible quadratic over $\mathbb{F}_2$ is $t^2 + t + 1$. If α is one of its roots, then $\alpha^3 = 1$, and

$$\overline{p}(\alpha) = \alpha^4 + \alpha^3 + \alpha^2 + \alpha + 1 = \alpha^2 \neq 0.$$

Thus $t^2 + t + 1$ does not divide $\overline{p}$, so $\overline{p}$ is irreducible. Gauss's lemma proves the claim. □

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3. THE SALEM CONJUGATES

For $t \neq 0$, put $u = t + t^{-1}$. Then

$$\frac{p(t)}{t^2} = u^2 - 3u + 1. \quad (3)$$

The two values of u are

$$u_+ = \frac{3 + \sqrt{5}}{2}, \quad u_- = \frac{3 - \sqrt{5}}{2},$$

and therefore

$$p(t) = (t^2 - u_+t + 1)(t^2 - u_-t + 1) \quad (4)$$

over $\mathbb{Q}(\sqrt{5})$.

Proposition 3.1. *The four roots of p are*

$$\beta, \quad \beta^{-1}, \quad z, \quad \bar{z},$$

where $\beta > 2$, $0 < \beta^{-1} < 1$, and $|z| = |\bar{z}| = 1$.

Proof. Since $2 < \sqrt{5} < 3$,

$$u_+ > \frac{5}{2} > 2, \quad 0 < u_- < \frac{1}{2}.$$

The first quadratic in (4) has two positive reciprocal real roots. At $t = 2$, its value is

$$4 - 2u_+ + 1 = 2 - \sqrt{5} < 0,$$

whereas its leading coefficient is positive. Its larger root β therefore satisfies $\beta > 2$, and the other is β^{-1} .

The discriminant of the second quadratic is negative because

$$u_-^2 - 4 < \frac{1}{4} - 4 < 0.$$

Its two roots are a nonreal conjugate pair whose product is one; hence each has modulus one. $\square$

By Lemma 2.1, p is the minimal polynomial of β . It is monic with integer coefficients, so β is an algebraic integer. Proposition 3.1 now proves that β is a Salem number in Schmidt's sense.

4. THE IRRATIONAL FIXED POINT

Set

$$x = \frac{1}{\beta - 1}.$$

Since $\beta > 2$, one has $0 < x < 1$. Moreover,

$$\beta x = \frac{\beta}{\beta - 1} = 1 + \frac{1}{\beta - 1} = 1 + x. \quad (5)$$

Reducing (5) modulo one gives

$$T_\beta(x) = x.$$

Thus the orbit of x is the singleton $\{x\}$, and $x \in \text{Per}(\beta)$.

The point is irrational. Indeed, if $x \in \mathbb{Q}$, then

$$\beta = 1 + \frac{1}{x} \in \mathbb{Q},$$

contradicting Lemma 2.1. This proves

$$x \in \text{Per}(\beta) \setminus (\mathbb{Q} \cap [0, 1)),$$

and hence proves Theorem 1.1. On the other hand, $x = 1/(\beta - 1) \in \mathbb{Q}(\beta)$, so the same point is compatible with Schmidt's open identity (2).

5. EXACT COMPUTATIONAL CHECK

The proof above is entirely symbolic. An accompanying exact-arithmetic check independently performs the following bounded tasks:

- (1) reduce p modulo two and exhaust the possible linear and irreducible quadratic factors;
- (2) verify the reciprocal identity (3) and reconstruct (4);
- (3) check the strict rational bounds used to obtain $\beta > 2$ and the unit-circle conjugate pair; and
- (4) verify the polynomial identity obtained by eliminating β from $x = 1/(\beta - 1)$, together with the fixed-point relation (5).

A separate typed polynomial-factor operation returns p as one monic factor and verifies exact product reconstruction. Since that operation does not itself certify factor irreducibility, the mathematical conclusion uses the explicit modulo-two proof in Lemma 2.1.

6. SCOPE AND REFEREE CHECKLIST

The result corrects only the right-hand side $\mathbb{Q}$ in the literal catalogue identity. It does not resolve or refute Schmidt's $\mathbb{Q}(\beta)$ problem. The distinction is substantive, not cosmetic: the counterexample point is itself an element of $\mathbb{Q}(\beta)$.

An external referee can check the conclusion by the following short list.

- (1) Verify irreducibility of $\bar{p}$ over $\mathbb{F}_2$.
- (2) Verify the substitution $u = t + t^{-1}$ and the two quadratic factors.
- (3) Check the real-root and unit-circle inequalities without numerical approximation.
- (4) Check $0 < x < 1$, $\beta x = 1 + x$, and the irrationality argument.
- (5) Compare the two distinct source formulations $\mathbb{Q} \cap [0, 1)$ and $\mathbb{Q}(\beta) \cap [0, 1)$.

REFERENCES

- [1] K. Schmidt, On periodic expansions of Pisot numbers and Salem numbers, *Bull. London Math. Soc.* **12** (1980), no. 4, 269–278.